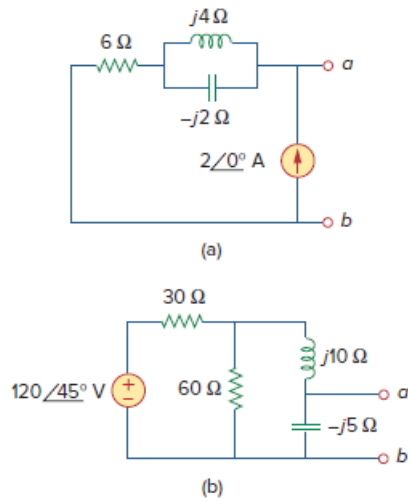


Problem

For each of the circuits in Fig. 10.99, obtain Thevenin and Norton equivalent circuits at terminals a - b .

Figure 10.99:



Step-by-step solution

Step 1 of 10

(a)

Refer to the Figure 10.99 (a) in the textbook.

Determine the impedance Z_{TH} .

Open circuit the current source.

The modified circuit diagram is shown in Figure 1.

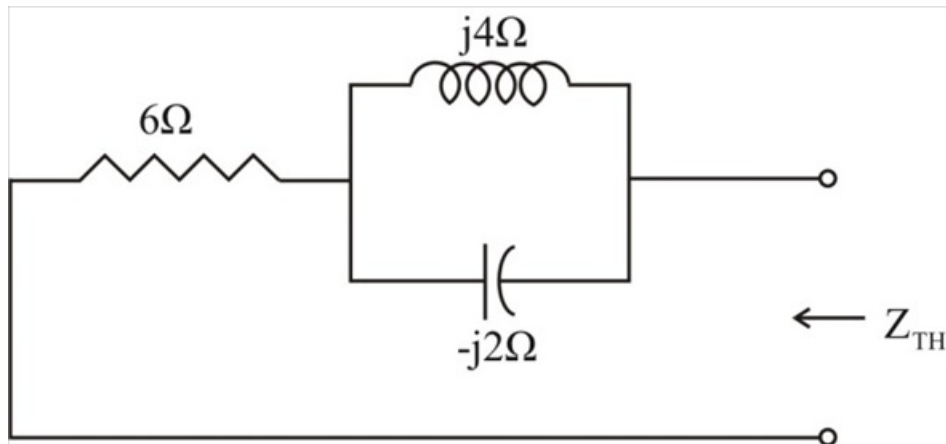


Figure 1.

The impedance Z_{TH} is,

$$\begin{aligned} Z_{TH} &= 6 + j4 \parallel (-j2) \\ &= 6 + \frac{(j4)(-j2)}{j4 - j2} \\ &= 6 - j4 \Omega \\ &= 7.211 \angle -33.69^\circ \Omega \end{aligned}$$

Since, the thevenin impedance and nortons impedance are equal.

Therefore, the equivalent impedance is $Z_N = Z_{TH} = 7.211 \angle -33.69^\circ \Omega$.

Step 2 of 10

Determine the nortons current I_N .

Short circuit the terminals a and b as shown in Figure 2.

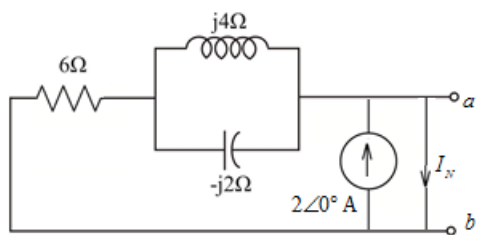


Figure 2

The norton's current I_N is equal to source current.

Therefore, the nortons current is $I_N = 2 \angle 0^\circ \text{ A}$.

Step 3 of 10

Determine the thevenin voltage.

$$V_{TH} = Z_{TH} I_N$$

Substitute $7.211 \angle -33.69^\circ \Omega$ for Z_{TH} , $2 \angle 0^\circ \text{ A}$ for I_N in equation.

$$\begin{aligned} V_{TH} &= (7.211 \angle -33.69^\circ)(2 \angle 0^\circ) \\ &= 14.422 \angle -33.69^\circ \text{ V} \end{aligned}$$

Therefore, the thevenin's voltage is, $V_{TH} = 14.422 \angle -33.69^\circ \text{ V}$.

Step 4 of 10

Norton's equivalent circuit is shown in Figure 3.

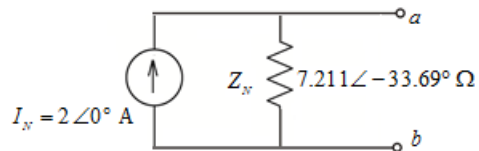


Figure 3

Thevenin's equivalent circuit is shown in Figure 4.

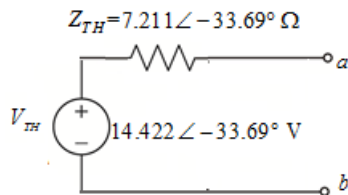


Figure 4

Step 5 of 10

(b)

Refer to the Figure 10.99 (b) in the textbook.

Determine the impedance Z_{TH} .

short circuit the all voltage sources.

The modified circuit diagram is shown in Figure 5.

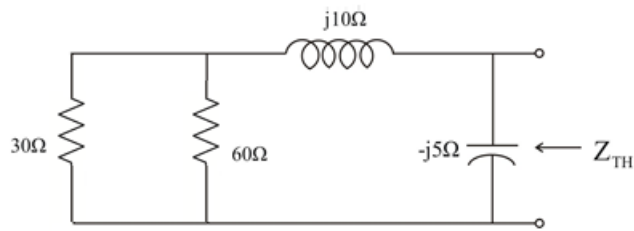


Figure 5.

The impedance Z_{TH} is,

$$\begin{aligned} Z_{TH} &= -j5 \parallel [(30 \parallel 60) + j10] \\ &= -j5 \parallel (20 + j10) \\ &= \frac{(-j5)(20 + j10)}{20 + j5} \\ &= 5.423 \angle -77.47^\circ \Omega \end{aligned}$$

Since, the thevenin impedance and nortons impedance are equal.

Therefore, the equivalent impedance is, $Z_N = Z_{TH} = 5.423 \angle -77.47^\circ \Omega$.

Step 6 of 10

Finding the open circuit voltage at terminals ab

$$\begin{aligned} V_{60\Omega} &= (120 \angle 45^\circ) \left[\frac{60 \parallel (j10 - j5)}{30 + (60 \parallel (j10 - j5))} \right] \\ &= (120 \angle 45^\circ) \left[\frac{60 \parallel j5}{30 + (60 \parallel j5)} \right] \end{aligned}$$

First we find the voltage across the 60 ohm resistor

$$\begin{aligned} &= (120 \angle 45^\circ) \left[\frac{\frac{60 \times j5}{60 + j5}}{30 + \frac{60 \times j5}{60 + j5}} \right] \\ &= (120 \angle 45^\circ) \left[\frac{j300}{1800 + j450} \right] \end{aligned}$$

$$\begin{aligned} V_{60\Omega} &= (120 \angle 45^\circ) \left[\frac{300 \angle 90^\circ}{1855.4 \angle 14.036^\circ} \right] \\ &= 19.4 \angle 120.964^\circ \text{ V} \end{aligned}$$

Step 7 of 10

Determine the thevenin voltage.

Apply voltage division rule.

$$\begin{aligned} V_{th} &= (19.4 \angle 120.964^\circ) \left(\frac{-j5}{j10 - j5} \right) \\ &= (19.4 \angle 120.964^\circ) \left(\frac{-j5}{j5} \right) \\ &= 19.4 \angle 300.964^\circ \text{ V} \end{aligned}$$

Therefore, the thevenin's voltage is, $V_{th} = 19.4 \angle -59.964^\circ \text{ V}$.

Step 8 of 10

Determine the nortons current I_N .

$$I_N = \frac{V_{TH}}{Z_{TH}}$$

Substitute $19.4 \angle -59.964^\circ \text{ V}$ for V_{TH} , $5.423 \angle -77.47^\circ \Omega$ for Z_{TH} in equation.

$$I_N = \frac{19.4 \angle -59.964^\circ \text{ V}}{5.423 \angle -77.47^\circ \Omega}$$

Therefore, the nortons current is, $I_N = 3.577 \angle 17.51^\circ \text{ A}$.

Step 9 of 10

Norton's equivalent circuit is shown in Figure 6.

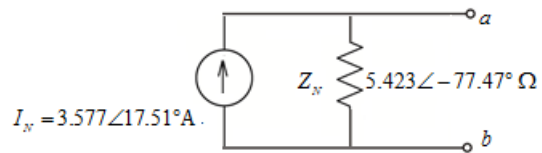


Figure 6

Step 10 of 10

Thevenin's equivalent circuit is shown in Figure 7.

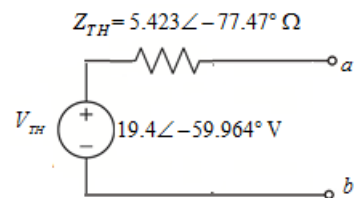


Figure 7